

# COMBINED ELECTRICITY AND WATER SUPPLY STRATEGY FOR THE DEPARTMENT OF WATER AFFAIRS

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## ABSTRACT

In the recent years various Demand Side Management (DSM) projects were successfully completed on Department of Water Affairs (DWA) pump schemes. Each pump scheme consists of a number of pump stations which transfer water to power stations in the Mpumalanga Highveld area and municipal off-takes.

The pump stations in a scheme are interconnected via rivers and pipe lines in order to transfer water from national key-point dams to different water users. Therefore, to manage the Highveld's water balance, all the interconnecting schemes work together to keep the important dams at a maximum operating level.

Introducing load management on these sites, local- and regional strategies were developed and rolled out. Local strategies involve upgrades to existing pump stations, while regional strategies involve the interactions of the different pump schemes. With the completion of the local strategies, regional strategies for combined water balance and energy optimisation can be developed.

By building a system that is more 'visible', data is automatically collected. This will then be used to develop an advanced management strategy for the DWA. Therefore, in this article an overview is given on completed load management projects, planned future projects and the strategy to develop a combined electricity and water supply strategy for the DWA.

## 1. INTRODUCTION

The Mpumalanga Highveld is the power house of large South African industries. It houses the chemical giant, Sasol, several coal mines and multiple power stations [1].

Water for these power stations is supplied by the Department of Water Affairs (DWA). This is because water in South Africa is considered a national asset for the benefit of all forms of life [2].

In the Highveld a number of GWSs deliver water to various clients. These clients include municipalities, farmers, mines, power stations. However, for the purpose of the study, the following interconnected water transfers schemes and pump stations are considered:

- **Usutu-Vaal scheme (Vaal scheme)**  
(Grootdraai/Tutuka, Grootfontein and Rietfontein pump stations)
- **Usutu scheme**  
(Jericho-, Kliphoek-, and Camden pump station)
- **Komati scheme**
- **Vaal River Eastern Subsystem Augmentation project (VRESAP) scheme**

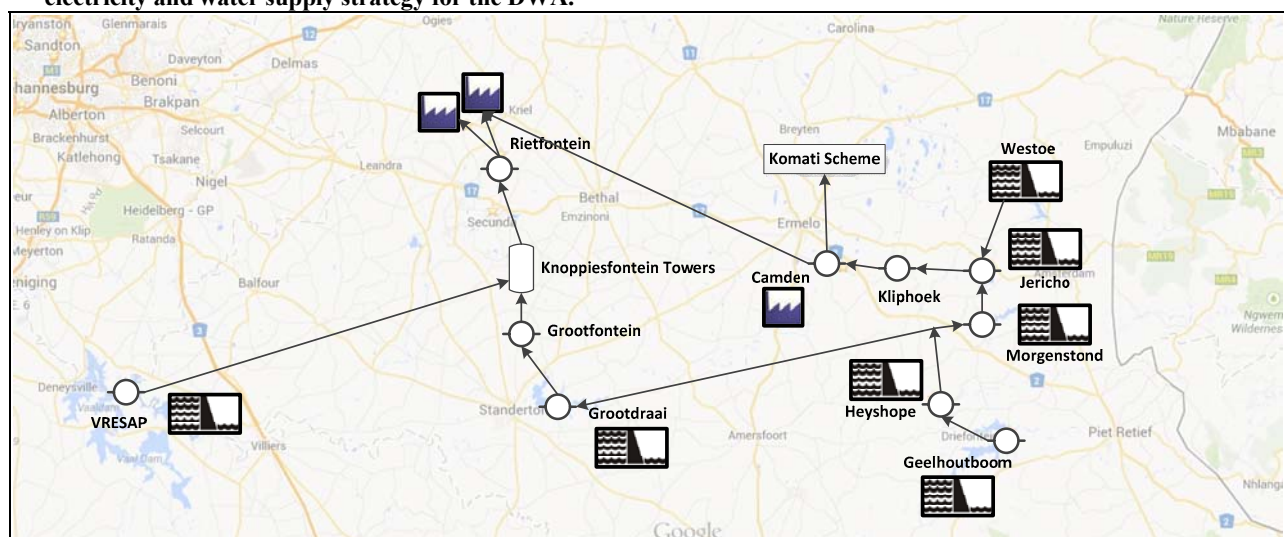


Figure 1: Pump station layout

Figure 1 represents a complex system of the four different GWSs which consist of dams, reservoirs and pump stations. The key dams are Grootdraai- (Vaal scheme) and Jericho dam (Usutu Scheme). These two dams are the

'starting-point' for transfers to the rest of the scheme. The other dams in Figure 1, augment these two vital dams:

- Geelhoutboom,
- Heyshope,
- Morgenstond,
- Westoe (gravity fed); and
- Vaal dam (VRESAP scheme).

VRESAP was completed in April 2009 in order to deliver water to the mothballed and new Eskom power stations, as well as to Sasol in Mpumalanga. This R2.7 billion pump station and 121 km pipeline project was implemented to meet the growing demand for water and as an assurance of supply for Eskom and Sasol for their respective power stations [3]. In this scheme water is transferred from the Vaal dam and ties in at the Knoppiesfontein towers of the Vaal scheme.

## 2. APPROACH IN DEVELOPING AN OPTIMISATION STRATEGY

In developing an energy optimisation strategy for the GWSs the approach in Figure 2 is followed. Therefore, by

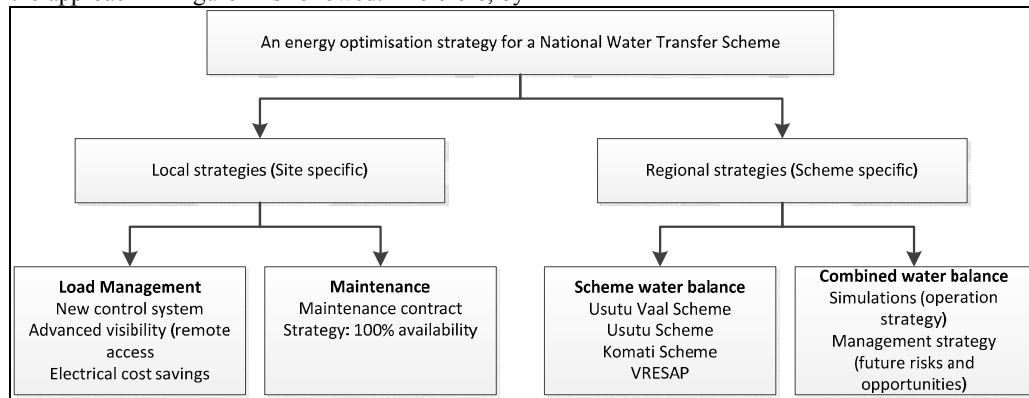


Figure 2: Approach to develop an optimised strategy

## 3. PUMP STATION CONTROL: DEVELOPING AN ADVANCED CONTROL SYSTEM

The scope of work for upgrading the pump stations included the removal of the old control systems; either old relay logic control panels or outdated Programmable Logic Controllers (PLCs). The upgrades included installation of:

- Compact Logix Ethernet Allen Bradley Processor (PLC) per pump-set.
- Single pump station PLC.
- Compact I/O Device Adapter per PLC.
- Human Machine Interface per pump station.
- RSLinx OEM SCADA software.
- Differential pressure transmitters for the dams/reservoirs.
- Communication to the medium voltage (MV) switchgear for the
  - Open circuit,
  - Close circuit; and
    - Trip circuit.
- Communication infrastructure to remote dam/reservoir and pump stations.

introducing energy management strategies on a national scale the first steps include:

- Local strategies and
- Regional strategies.

Local strategies involve physical site specific upgrades. This is done because most of DWA's pump stations are advanced in years and require equipment and control system upgrades.

With the completion of the local strategies, regional strategies for combined water balance and energy optimisation can be developed. This strategy should consider the interaction of the different GWS, by also including potential risks and future opportunities. Therefore, a regional model needs to be created to aid in building this strategy.

Installations on all six pump stations were done according to Rockwell's international standards and the design of all the pump-sets are similar to each other. For the design, the analogue and digital signals of the 30 pump-sets are 95% similar, depending what instrumentation is not installed for a specific pump-set. These measurements included:

- Bearing temperatures,
- Winding temperatures,
- Pump casing temperature,
- Main pipeline pressures and flows,
- Oil pressure switches and flows,
- Vibrations, and
- Power measurements: Voltage, ampere and power factor.

The analogue and digital signals from a pump-set are used in the SCADA system for continual condition monitoring. When an unhealthy signal is measured, the pump-set will automatically be stopped. Only when the fault has been rectified in the field, will it be able to start the pump-set. A SCADA presentation of a pump-set can

be seen in Figure 3. The following control modes are available:

- Automatic pump control,
- Manual control; and
- Test mode. For testing purposes for valves and motors.

With the completion of installations at a pump station, the control philosophy of the station is programmed into the PLC. By the press of a single button a pump or even the entire pump station can be operated. The updated control philosophy includes the scheme's pump

operations but also incorporate load management. Advantages for the projects include:

- Hardware upgrades with up-to-date PLC technology,
- Full graphic display SCADA system,
- Full condition monitoring of the pumps,
- Data logging and trending for each pump,
- Advanced control philosophy,
- Remote dam level monitoring; and
- Password protected remote monitoring of the pump station which is restricted to specific authorized DWA personnel.

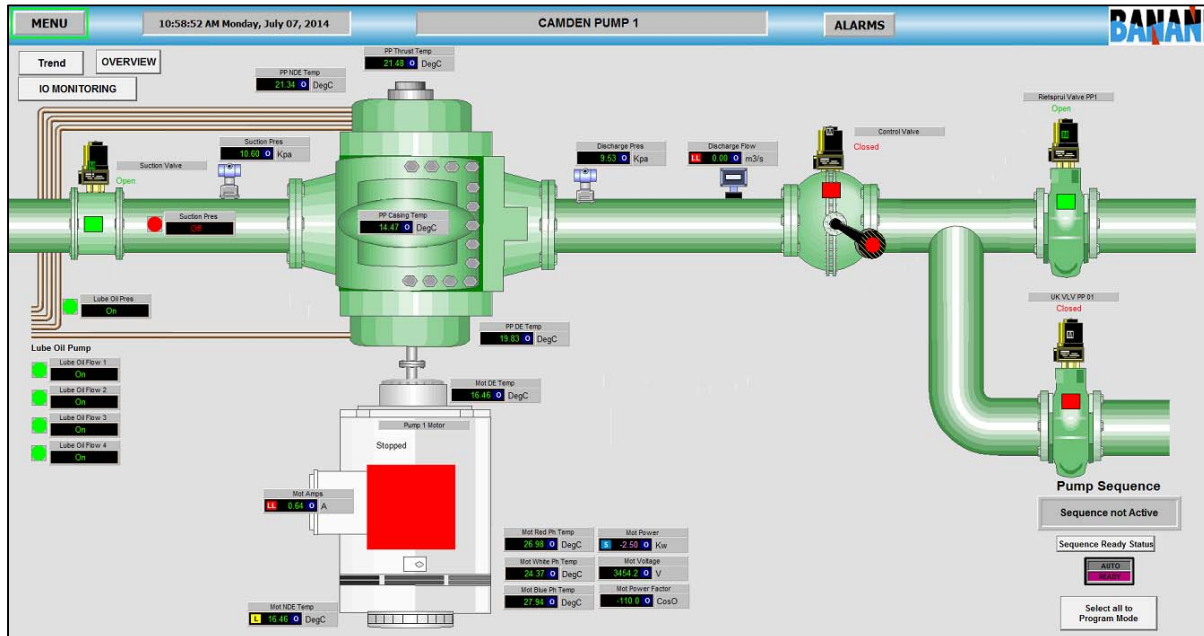


Figure 3: SCADA preview: Camden pump 1

#### 4. LOCAL STRATEGY / LOAD MANAGEMENT PROJECTS

In the last three years load energy peak clip projects have been done on the Vaal- and Usutu schemes. By looking at the performances of these projects, the interactions of the schemes will be highlighted in the preceding sections. Table 1 show the project dates of the mentioned GWS.

Table 1: Project dates

| Action taken                    | Phase 1:<br>Vaal | Phase 2:<br>Usutu |
|---------------------------------|------------------|-------------------|
| Order placement                 | 05-Sep-11        | 10-Jul-13         |
| Project installation completion | 31-Jan-12        | 31-Mar-14         |

|                               |                        |                        |
|-------------------------------|------------------------|------------------------|
| Performance assessment period | March 2012 to May 2012 | July 2014 to Sept 2014 |
|-------------------------------|------------------------|------------------------|

#### 5. PHASE 1: VAAL PROJECT

During 2011 the first investigations started for a DSM load shift project on the Vaal GWS. The layout of the scheme can be seen in Figure 4. The distance above sea level is indicated in the figure. The first set of upgrades was done on the:

- Grootdraai/Tutuka (GD/TK)-,
- Grootfontein (GF)- and
- Rietfontein (RF)-pump stations.

The upgrades mentioned in section 3 were implemented at the pump stations. A presentation of an upgraded control panel is shown in Figure 5.

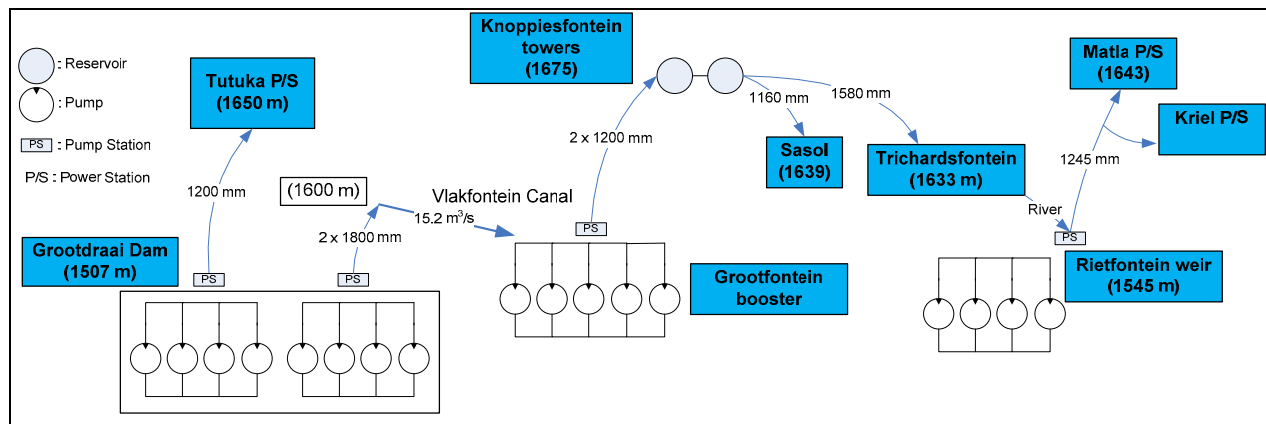


Figure 4: Vaal GWS



Figure 5: Upgraded control panel at Grootdraai

#### Grootdraai/Tutuka

The pump station consists of eight pump sets, 4 pumps-sets transfer water to Grootfontein, while the other 4 pump-sets are responsible for providing water to Tutuka power station. Grootdraai pump station consists of four 1.65MW induction motor pump-sets. The Tutuka part consists of a 0.32 MW booster- and a 1.4 MW main pump per pump-set.



Figure 6: Grootdraai pump-set 1

A major storage dam was constructed at Grootdraai near Standerton from where water is transferred via a 41km canal and pipelines towards Grootfontein. A Grootdraai pump is presented in Figure 6.

#### Grootfontein

The GF pump consists of one 2.15 MW and one 2.2 MW induction motor pump-sets. GF pump station serves as a booster pump station. Water enters GF from the endpoint of the canal (fore bay) and is pumped to Knoppiesfontein towers. These reservoirs are the highest points above sea level; from here the water is gravity fed to Trichardsfontein dam and also Sasol II and III.



Figure 7: Grootfontein pump station

#### Rietfontein

From Trichardsfontein water and via rivers water end at Rietfontein dam. From this dam water is transferred to Matla and Kriel power stations. Rietfontein consist of four 3.05 MW induction motor pump-sets.

Rietfontein pump station is shown in Figure 8.



Figure 8: Rietfontein pump station

Through various project challenges, this project was successfully completed by May 2012 and delivered a total load shift saving of 13.40 MW [4], [5]. The results are presented in Figure 9. This shows that the load was shifted out of the Eskom evening peak of 18h00 to 20h00.

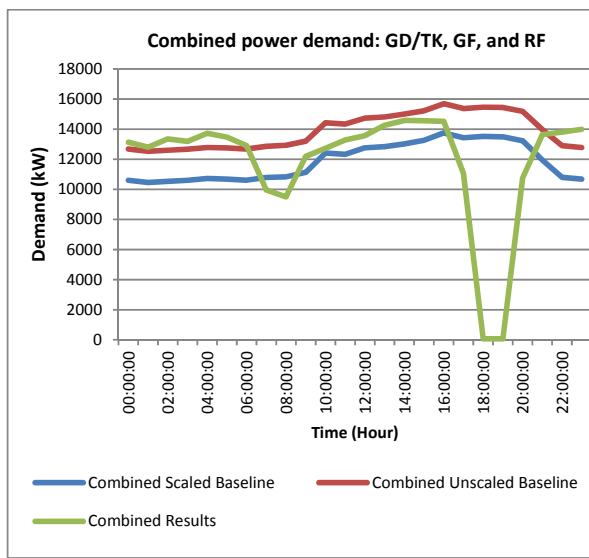


Figure 9: Vaal project savings

## 6. PHASE 2: USUTU PROJECT

A similar project was undertaken in 2013 on the Usutu GWS pump stations:

- Jericho-,
- Kliphoek-, and
- Camden pump stations.

### Jericho pump station:

Jericho pump station consists of four induction- (3.455 MW) and two Synchronous motor (3.65 MW) pump-sets. Water is withdrawn from Jericho dam and is pumped west towards Kliphoek pump station through two pipelines, called the north- and south pipeline.



Figure 10: Jericho pump station

Jericho pumps 1 to 3 are allocated to the South line, while pumps 4 to 6 pump through the North line. However, the two Synchronous motor pump-sets (3 and 6) are interchangeable on the two pipe lines, thus adding to the redundancy of the pump station.



Figure 11: Jericho control panel modifications

### Kliphoek pump station:

Kliphoek pump station, as seen in Figure 12, only acts as a booster pump station and consist of four 2.4 MW induction motor pump-sets. The North and South pipelines run directly underneath the pump station. A pump will only be added per pipeline when additional flow is required due to a higher water demand. Only one pump is allowed to be added per pipeline and also only when two Jericho pump are already pumping through a specific pipeline.

A suction pressure transmitter monitors the 'number of pumps' running at Jericho. Therefore, when a lower pressure is recorded, a stop signal will be sent to the pump-set, indicating that one or two Jericho pumps have been stopped. However, during a normal stop procedure, the control operator will first stop the Kliphoek pump before a Jericho pump will be stopped.



**Figure 12:** Kliphoek pump station

From Kliphoek water is pumped towards the Onverwacht reservoirs from which water is then gravity fed to the Camden reservoirs 1 through 3.

#### Camden pump station

Camden pump station is situated on Eskom grounds at the power station and consists of three 2.8 MW induction motor pump-sets. Eskom uses the water from reservoir 1 and 2 while the DWA pumps water from reservoir 3 to Rietspruit reservoirs to be used by the Ermelo municipality

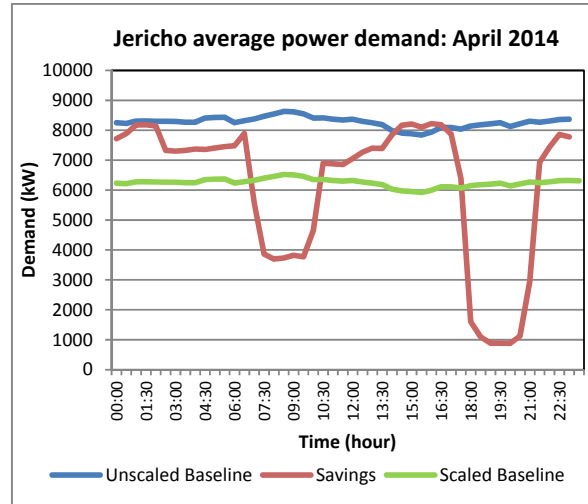
From Camden pump station water can also be pumped to the Upper Komati (UK) scheme.

Both the Kliphoek and Camden pump stations are unmanned pump stations and are controlled remotely from the control room situated at Jericho dam.

After successful the installation and commissioning of the pump control philosophy, the Usutu project entered into the performance assessment period. However, at the time of writing, only the results of April 2014 were available.

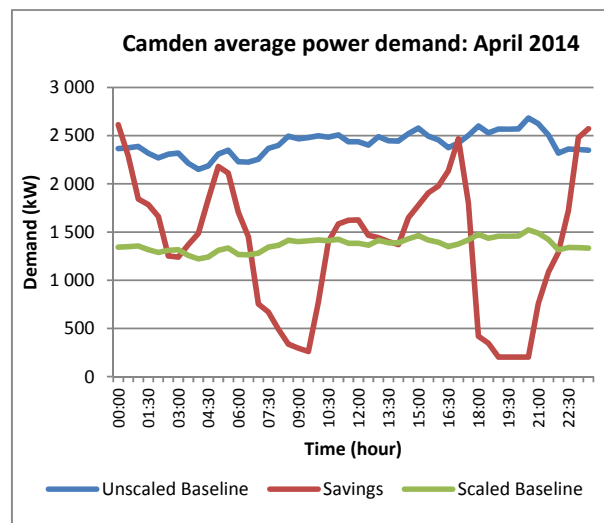
Before the start of the project a power baseline was recorded. Therefore, during the performance assessment period this baseline is scaled up or down, depending on the change in demand, as seen in Figure 13 and Figure 14. The initial baseline will be scaled up or down depending on the performance assessment month power profile.

Therefore, the load shift savings achieved at Jericho pump station for April 2014 were 3.74 MW. Therefore, 93.5% of the target savings (4.0 MW) were achieved. The results are plotted in Figure 13.



**Figure 13:** Jericho savings

However, the load shift savings achieved at Camden pump station was only 930 kW. This was only 40% of the target savings (2.33 MW). The savings can be seen in Figure 14.



**Figure 14:** Camden savings

Almost no pumping occurred at the Kliphoek pump station; therefore no savings have been recorded.

Every year DWA revises the national operational rules, which are called May One rules. This set of operational rules lays out the principles for water transfers between GWSs, the volumes that need to be transferred, the operating levels of the national dams, risk assessments and how the national water balance should be managed. Therefore, during the implementation of the project, the operational rules for the two GWSs changed. This explains the under performance and therefore needs further investigation.

## 7. INTEGRATED SCHEME WATER BALANCE, INCORPORATING LOAD MANAGEMENT

From Figure 13 and Figure 14 it is clear that the current pump operations are much lower than the original recorded baseline.

Before the start of the project the Usutu scheme operated at a higher demand, delivering 75% of the water to Matla and Kriel power stations, while the Vaal scheme was responsible for the remaining 25% (Note Figure 16). However, due to high strain on the Usutu scheme to meet the systems water demands, the Vaal scheme became responsible for 75% of the water and the Usutu scheme for the remaining 25%.

The water demand on the Usutu scheme increased during 2013 when a strategically important dam in the Komati GWS dropped to an extremely low operating level (+49%). This dam, Nooitgedagt, also transfers water to the Eskom power stations and is augmented from Camden pump station in the Usutu Scheme.

The interactions of these schemes can be seen in Figure 16, which shows the Usutu GWS and the Vaal GWS, as well as Rietfontein pump station. Rietfontein in return, is augmented from Grootdraai/Tutuka-, Grootfontein- and VRESAP pump stations.

Therefore, due to the change in operating rules, the pump operations at Rietfontein rapidly increased. As seen in Figure 15, the baseline increased by 1.9 times.

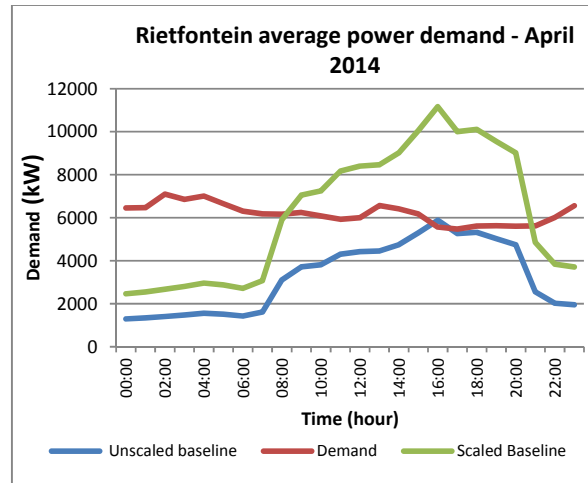


Figure 15: Rietfontein increased demand

Therefore, although minimal load shifting was done, the savings achieved at Rietfontein for April 2014 was 4.31 MW with a target of 3.2 MW. A performance of 134%.

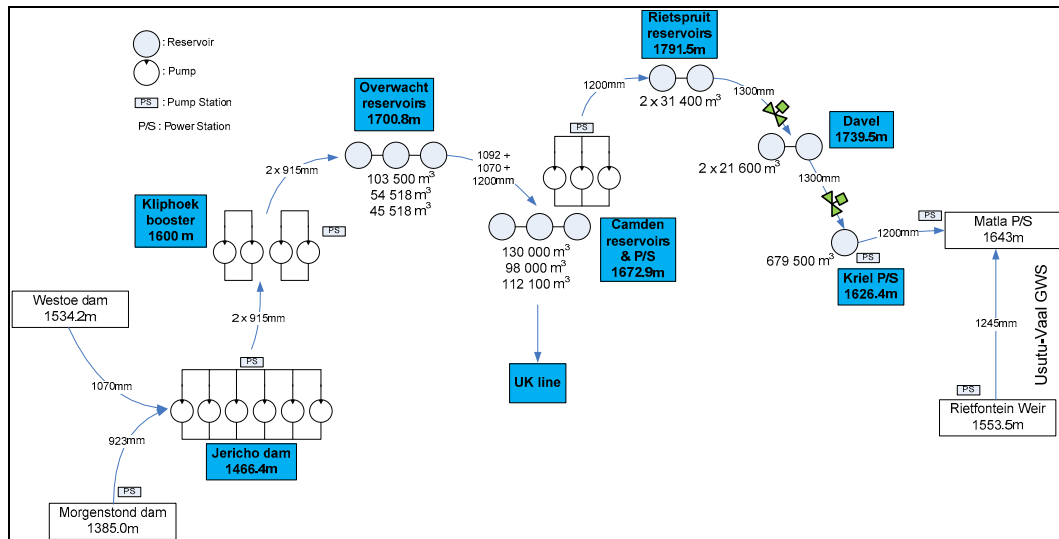


Figure 16: Usutu GWS and Rietfontein pump station (Vaal Scheme)

## 8. CONCLUSION AND FUTURE WORK

During the last couple of years the operational rules for the Vaal and Usutu schemes changed to accommodate the increased demand. This had a significant impact on the performance on the various pump schemes.

For the correct savings to be presented by each pump scheme, the baselines of the Vaal and Usutu schemes should not only be combined, but the regional water transfers should also be included into the calculations.

Considering the combined savings of the Vaal and Usutu schemes, a total of 9 MW savings was achieved. This represented a performance factor of 76%.

However, during this time the operators at Rietfontein pump station were exempted from doing load shifting. With proper planning, maintenance and an integrated view it will be possible to shift 9 MW out of the Eskom evening peak.

A SCADA presentation is shown in Figure 17. This page is being developed to assist in building a centralised monitoring station for the regional water transfers. This system will be able to provide the statuses of the specific

pump-sets in a scheme; indicate the remote dam/reservoir levels; and measure the current flow at specific pipe sections. A SCADA page is being developed for the Vaal and interconnected schemes. This will be monitored from DWA headquarters in Gauteng.

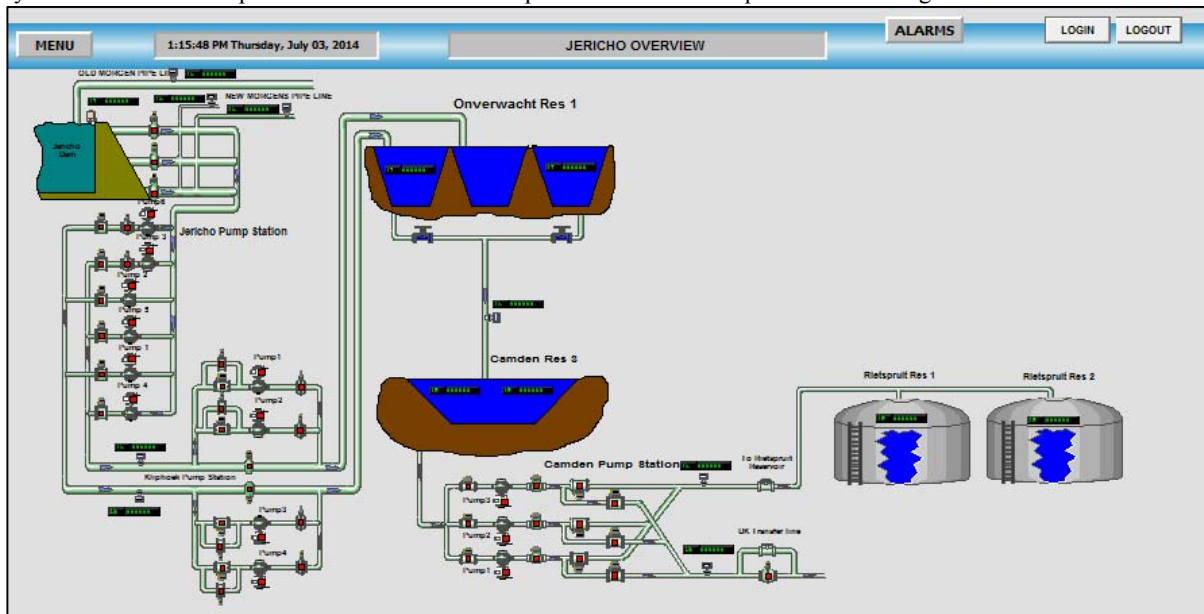


Figure 17: SCADA Design - Usutu remote monitoring

Up to the present, only 50% of the DWA pump-stations in the Mpumalanga Highveld were upgraded as part of the process to develop an electricity and water supply strategy for the DWA. Therefore, to include all the interconnected GWSs, future investigations, upgrades and work are planned for:

- Komati and
- VRESAP scheme.

These pump schemes directly influence the Vaal and Usutu Schemes. Therefore, future work also involves investigating other augmenting pump stations, which include:

- Geelhoudboom,
- Heyhope; and
- Morgenstond.

By including all the interconnected GWSs in the Mpumalanga Highveld, a complete model can be built to develop an integrated electricity and water supply strategy.

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